PANJAB UNIVERSITY, CHANDIGARH | UIET | ECE, Section 2

PROJECT SYNOPSIS

FPGA-Accelerated Lane Detection with Confidence-Controlled Edge Thresholding, Virtual-Lane Fallback and Speed-Breaker Warning for Indian Roads Using PYNQ-Z2

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1. Abstract and Problem Statement

This project proposes a lane detection prototype on the PYNQ-Z2 (Zynq-7000) board for Indian road conditions, where lane paint is often faded, shadowed or missing. Most lane detectors only report "lane found" or "not found". Our system adds four planned features: a marking-quality score, a score-controlled adaptive edge threshold in the FPGA, a virtual-lane fallback for unmarked roads, and a painted speed-breaker warning. A CPU-only version of the same algorithm is the baseline for comparing latency, FPS, accuracy and resource use. This synopsis states plans only; no measured result is claimed.

2. Objectives

1. Build a CPU lane detection baseline in Python (OpenCV, NumPy).
2. Build FPGA blocks for grayscale, Gaussian filter, Sobel edge detection and adaptive threshold, connected through AXI DMA and AXI4-Stream.
3. Compute a per-frame marking-quality score and use it to control the FPGA threshold at run time.
4. Estimate a virtual centre line when markings are missing, and detect painted speed breakers with a warning.
5. Compare CPU and FPGA versions, and fixed versus score-controlled thresholds, on a small hand-labelled Indian road clip set.

3. Proposed Features (What We Will Do)

No.	Feature	What it does	Runs on
1	Marking-quality score	Gives each frame a score from 0 to 1 using coverage (rows with edge support), edge strength and curve-fit error. Low score means faded or missing paint. Scores are logged to give a marking-condition profile of each clip.	Python (ARM)
3	Score-controlled adaptive threshold	The FPGA sets the edge threshold from the local average gradient (line buffer and running sum). The score from feature 1 is written back through an AXI-Lite control register, so the threshold adapts live to shadow, glare and faded paint.	FPGA + Python
4	Virtual-lane fallback	When the score stays low, the road surface is grown from a bottom-centre seed using colour (HSV) and texture, and a virtual centre line is drawn from the road boundaries. It is labelled "virtual".	Python (ARM)
2	Speed-breaker warning	Looks for repeated strong horizontal edge bands in the road region (painted speed breakers) and checks stripe spacing and width to reject zebra crossings and shadows. Raises "speed breaker ahead".	Python (ARM), reuses Sobel output

Processing flow: video frame, then FPGA (grayscale, Gaussian, Sobel, adaptive threshold), then Python (lane detection and fitting, quality score, virtual lane if score is low, speed-breaker check, lane-departure warning, overlay). The score is fed back to the FPGA threshold register.

4. Components Required

Hardware	Role
PYNQ-Z2 board (XC7Z020, dual-core ARM Cortex-A9, 53,200 LUT, 106,400 FF, 140 BRAM, 220 DSP, 512 MB DDR3)	Runs the FPGA blocks and the Python stages
Laptop	Vivado builds, Jupyter access, optional CPU reference run
Mobile phone camera	Records Indian road clips (video files are transferred to the board)
microSD card (16 GB or more), Ethernet cable	PYNQ image and connection to Jupyter
USB power meter (optional)	Energy per frame, CPU versus FPGA

Software	Use
Python, Jupyter, PYNQ	Overlay loading, DMA control, buffers, evaluation scripts
OpenCV, NumPy	CPU baseline, fitting, scoring, virtual lane, speed-breaker logic
Vivado, Verilog or VHDL	FPGA blocks, AXI DMA and AXI4-Stream design, bitstream, utilization report
Annotation tool (for example LabelMe)	Hand-labelling lanes, road centre and speed breakers

5. Methodology (Summary)

1. Record and hand-label Indian road clips: clear paint, faded paint, shadow and glare, unmarked road, traffic, painted speed breakers and zebra crossings.
2. Implement the CPU baseline: region of interest, grayscale, Gaussian, Sobel, lane detection (Hough or sliding window), curve fitting.
3. Move grayscale, Gaussian, Sobel and the adaptive threshold into the FPGA; check outputs against the CPU version.
4. Add the quality score, then the threshold feedback register, then the virtual lane, then the speed-breaker detector.
5. Add lane-departure warning and overlay. Curvature and a short-range trajectory (10 to 50 m) are attempted only where camera calibration allows.
6. Run both versions on the same clips, repeat runs, exclude warm-up frames, and report mean and spread.

6. Evaluation Plan

Metric	How it is checked
Latency (ms per frame), FPS, speed-up	Timestamps in code, CPU versus FPGA, per stage and full pipeline
Lane detection accuracy	Hand-labelled frames, pixel tolerance and overlap
Fixed versus score-controlled threshold	False-edge count and detection accuracy on shadow, glare and faded clips

Metric	How it is checked
Score validity	Low score versus hand-labelled faded frames (correct, missed, false flags)
Virtual-lane error	Lateral difference from annotated road centre on unmarked clips
Speed-breaker warning	Correct, missed and false warnings, including zebra-crossing and shadow clips
FPGA resource use; power (optional)	LUT, FF, BRAM, DSP from the Vivado report; power only if reliably measurable

Failure cases will be reported, not removed.

7. Novelty and Contribution

The individual techniques (lane confidence, adaptive thresholding, unmarked-road detection, vision-based speed-bump detection) exist in the literature. To our knowledge, the contribution here is their integration into one confidence-gated pipeline in which the quality score controls the FPGA threshold at run time, with a virtual-lane fallback and speed-breaker warning, evaluated on a labelled set of Indian faded, shadowed and unmarked roads with a CPU versus FPGA comparison on the PYNQ-Z2. A literature search will be documented in the final report.

8. Limitations and Scope

- Prototype evaluated on recorded video only; not for use in a vehicle. Out of scope: object and pedestrian detection, live capture (future work), safety certification.
- The virtual lane depends on road-surface colour and may fail with dust, tar patches or heavy shadow. Speed-breaker detection works only on painted speed breakers.
- The 10 to 50 m range, prediction accuracy and speed-up are unknown until measured. PYNQ-Z2 resources limit filter size and resolution.

9. Key References

[1] PYNQ documentation (DMA library), pynq.readthedocs.io. [2] TUL, PYNQ-Z2 Reference Manual v1.1. [3] J. Canny, "A computational approach to edge detection," IEEE TPAMI, 1986. [4] R. O. Duda and P. E. Hart, "Use of the Hough transformation to detect lines and curves in pictures," CACM, 1972. [5] G. Varma et al., "IDD: a dataset for exploring problems of autonomous navigation in unconstrained environments," WACV, 2019. [6] AMD (Xilinx), AXI DMA LogiCORE IP Product Guide, PG021.